

Part I

①

$$V = 30 \text{ V}$$

$$I = 200 \text{ A}$$

$$\text{Area of joint} = 20 \text{ mm}^2 = 20 \times 10^{-6} \text{ m}^2 = A$$

$$\text{weld speed} = 5 \text{ mm/s} = 5 \times 10^{-3} \text{ m/s}$$

$$\text{Heat required} = 20 \text{ J/mm}^3$$

$$\begin{aligned} \rightarrow \text{Power} &= \text{heat required per sec in welding} \\ &= \text{Heat required} \times A \times \text{weld speed} \\ &= 20 \times 5 \times 20 = 2 \text{ KJ/s} \end{aligned}$$

$$\begin{aligned} \rightarrow \text{Power supplied} &= \cancel{30} V \times I \\ &= 30 \times 200 = 6000 \Rightarrow 6 \text{ KJ/s} \end{aligned}$$

$$\text{Say time of welding} = t$$

$$\therefore \text{Heat supplied} = 6t \text{ KJ}$$

$$\text{Heat used} = 2t \text{ KJ}$$

$$\text{Heat lost} = 4t \text{ KJ}$$

$$\% \text{ heat lost} = \frac{4t}{6t} \times 100 = \frac{200}{3} = \boxed{66.67 \%}$$

②

$$V_{oc} = 100 \text{ V}$$

$$I_{sc} = 2000 \text{ A}$$

$$V = 10 + 5L \quad (\text{where } L \text{ is arc length in mm})$$

for Power max.

$$V_{max} = \frac{V_{oc}}{2} = 50 \text{ V}$$

$$I_{max} = \frac{I_{sc}}{2} = 500 \text{ A}$$

$$\begin{aligned} \therefore P_{max} &= V_{max} \times I_{max} = 50 \times 500 \\ &= \boxed{25 \text{ KVA}} \end{aligned}$$

(3)

thickness = to

$$V = 20 \text{ V}$$

$$I = 150 \text{ A}$$

$$\text{Welding speed} = 5 \text{ mm/s}$$

$$\text{Filler wire [diameter} = 3 \text{ mm / radius} = 1.5 \text{ mm]}$$

L

$$\text{Say feed rate} = f \text{ mm/s}$$

→ weld bead Composition

0.6 times of filler

0.4 times of plate

$$\rightarrow \text{Heat required} = 10 \text{ J/mm}^3$$

$$\rightarrow \text{Heat supplied} = 20 \times 150 = 3 \text{ kJ/s}$$

$$\rightarrow \text{Actual heat input (to weld)} = 3 \text{ kJ/s} \times 0.7 \times 0.6$$

$$= 1.26 \text{ kJ/s}$$

Say total ^{cross section} filled is $A \text{ mm}^2$ = ~~0.6~~ Vol filled by filler per sec

$$A \times 5 \times 0.6 = f \times \pi \times \left(\frac{3}{2}\right)^2$$

$$3A = 7.07 f \quad \text{--- (1)}$$

~~Vol filled by plate per sec~~

$$\text{--- } A \times 5 \times 0.4 \text{ ---}$$

Heat balancing

$$\text{Heat input} = \text{heat used}$$

$$1.26 \text{ kJ/s} = A \times 5 \times 10 \text{ J/s}$$

$$\Rightarrow 1.26 \times 1000 = A(50)$$

$$A = \frac{1260}{50} = 25.2 \text{ mm}^2$$

$$\text{Cross section filled} = 25.2 \text{ mm}^2$$

using

$$f = \frac{3 \times 25.2}{7.07} = 10.69$$

$\therefore$ feed rate of filler wire in mm/s.

$$04 = 10.7 \text{ mm/s}$$

(4)

$V = 100 + 40L$, where L is length of arc in mm
 V is voltage of arc

$$L_1 = 1.0 \text{ mm} \quad \& \quad L_2 = 2.0 \text{ mm}$$

$$I_1 = 250 \text{ A} \quad \& \quad I_2 = 200 \text{ A}$$

$$\therefore V_1 = 100 + 40(L_1) = 140 \text{ V}$$

$$V_2 = 100 + 40(L_2) = 180 \text{ V}$$

from eq:-

$$\frac{V_{oc} + I_{sc}}{V_{oc}} + \frac{I}{I_{sc}} = 1$$

$$\text{Say } \frac{1}{V_{oc}} = x \quad \& \quad \frac{1}{I_{sc}} = y$$

$$\therefore \textcircled{i} \quad 140x + 250y = 1$$

$$\textcircled{ii} \quad 180x + 200y = 1$$

$$\Rightarrow 5(180x + 200y) - 4(140x + 250y) = 1$$

$$\Rightarrow (900 - 560)x = 1 \Rightarrow 340x = 1$$

$$x = \frac{1}{340} = \frac{1}{V_{oc}}$$

$$\therefore V_{oc} = 340 = \text{Open circuit Voltage}$$

$$140\left(\frac{1}{340}\right) + 250y = 1 \quad \& \quad y = \frac{0.59}{250} = \frac{1}{425}$$

$$\therefore y = \frac{1}{425} = \frac{1}{I_{sc}}$$

$$\therefore \text{Short Circuit Current} = I_{sc} = 425 \text{ Amp}$$